

**Retail Flow-Induced Gamma Dislocation:
A Dealer-Hedging Theory of Short-Horizon
Equity Return Predictability and Its Systematic
Exploitation**

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Abstract

We formalize and extend the empirical findings of Barclays Equity Derivatives Strategy’s “Impact of Retail Options Trading” (Deshpande et al., 2020), which documents a threefold year-on-year increase in U.S. single-stock option volume following the October 2019 elimination of retail brokerage commissions, concentrated in short-dated (sub-two-week) call options on large-capitalization technology and e-commerce equities. We derive, from a Kyle/Grossman-Miller inventory-based market-making model, the precise mechanism by which uninformed but sufficiently large retail call-buying flow forces dealers into delta-hedging purchases of the underlying, generating a measurable and exploitable short-horizon return-predictability signal. We construct two candidate factors — Excess Option Volume (EOV) and Small-Lot Call Skew (SLCS) — and show, replicating the source report’s cross-sectional regression methodology, a regime shift in predictive power coincident with the zero-commission announcement. We then present a complete systematic trading system: signal construction, point-in-time backtesting with transaction-cost and borrow-cost modeling, mean-variance portfolio construction with turnover penalties, fractional-Kelly position sizing, a gamma-squeeze tail-risk circuit breaker, and a production monitoring and signal-decay detection framework. We situate the strategy within the post-2020 academic literature on option-demand-based asset pricing and retail options trading, arguing the underlying mechanism is structural rather than a pandemic-era artifact.

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Chapter 1

Introduction

1.1 Motivation

The market microstructure literature has long recognized that order flow, even when uninformed in the traditional sense, can move prices through inventory and hedging channels rather than information channels (1; 2). The 2019–2020 zero-commission shock to U.S. retail brokerages (Charles Schwab, October 2, 2019, followed immediately by TD Ameritrade, E*Trade, and Fidelity) produced a natural experiment: a discrete, dateable, exogenous change in the marginal cost of trading options that Barclays Equity Derivatives Strategy documents produced a threefold increase in single-stock option volume, concentrated in short-dated calls.

1.2 Contribution

This dissertation makes four contributions beyond the original sell-side research note:

1. A formal first-principles derivation of the dealer-hedging price-impact mechanism (Chapter 2), connecting the empirical regularities to the Kyle (1985) and Grossman–Miller (1988) inventory models and to Gârleanu, Pedersen & Poteshman’s (2009) demand-based option pricing framework.
2. A literature review situating the finding within the post-2020 academic record (Chapter 3), establishing that the phenomenon is persistent rather than a COVID-era artifact.
3. A complete, cost-realistic, walk-forward backtesting and portfolio-construction framework (Chapters 4–6) suitable for institutional evaluation.
4. A production monitoring and signal-decay detection specification (Chapter 7) addressing the central risk to any flow-based strategy: crowding and alpha decay.

Chapter 2

Theoretical Framework

2.1 The Dealer Inventory Problem

Consider a representative options market maker who, over trading interval t , receives net customer call-buying order flow $Q_{i,t}^c$ in single name i . In the Grossman–Miller (1988) framework, the market maker is a risk-averse intermediary who will only hold inventory imbalances temporarily, hedging them in the correlated (underlying equity) market. Absent full two-sided offsetting flow, the dealer’s residual delta exposure requires a hedge trade in the underlying of

$$\Delta H_{i,t} = N_{i,t} \cdot m \cdot \Delta_{c,i,t} \cdot (1 - \phi_{i,t}) \quad (2.1)$$

where $N_{i,t}$ is net contracts sold to customers, $m = 100$ is the contract multiplier, $\Delta_{c,i,t} \in (0, 1)$ is the Black–Scholes call delta, and $\phi_{i,t} \in [0, 1]$ is the fraction of the order absorbed by natural two-way flow.

2.2 Price Impact and the Feedback Loop

Under a linear price-impact assumption (Kyle, 1985), the dealer’s hedge trade moves the price by

$$\Delta P_{i,t} = \lambda_{i,t} \cdot \Delta H_{i,t} \quad (2.2)$$

where $\lambda_{i,t}$ is the (time-varying, liquidity-dependent) Kyle-lambda price-impact coefficient. Critically, because retail call-buying is documented (Barclays Fig. 5, 16) to be persistent and directionally one-sided — net call buy-to-open volume roughly doubling relative to sell-to-open post-regime-switch, while institutional flow stayed range-bound — the hedge-driven price impact is not immediately offset by mean-reverting flow, producing a short-horizon *momentum* signature rather than the typical liquidity-provision mean-reversion signature. This is the reflexive loop:

Call buying $\rightarrow$ Dealer short gamma $\rightarrow$ Dealer buys stock $\rightarrow$ Price $\uparrow \rightarrow$ More call buying (extrapolative retail demand)

2.3 Demand-Based Option Pricing and the Volatility Surface

Gârleanu, Pedersen & Poteshman (2009) show theoretically that when end-user option demand cannot be fully hedged by risk-neutral arbitrageurs (due to jump and volatility risk that cannot be perfectly replicated), net demand distorts the implied volatility surface in the direction of the

demand imbalance. This precisely predicts the two volatility-surface findings independently documented by Barclays: (i) skew flattening in high-retail-flow names (excess demand for calls relative to puts lowers relative put pricing pressure, flattening skew — Fig. 17), and (ii) a muted volatility risk premium in the same names despite elevated implied volatility, because realized volatility itself rises in tandem via the hedging feedback loop (Fig. 19–20) — i.e., the “excess” implied volatility is not pure premium, it is compensation for genuinely higher subsequent realized volatility induced by the flow itself.

Chapter 3

Literature Review

- **Kyle (1985)**: foundational linear price-impact model; provides λ in our framework.
- **Grossman & Miller (1988)**: inventory-based market-making; motivates the dealer hedge-timing assumption.
- **Gârleanu, Pedersen & Poteshman (2009, *Review of Financial Studies*)**: demand-based option pricing; theoretical basis for skew/VRP distortion.
- **Ni, Pearson, Poteshman & White (2021, *Journal of Financial Economics*)**: documents causal price effects of options-related hedge rebalancing around expiration, directly corroborating the dealer-hedging channel with a cleaner identification strategy than volume correlations alone.
- **Barbon & Buraschi (2022 working paper)**: full-book dealer gamma imbalance forecasts realized volatility; motivates our bipower-variation realized-vol estimator and gamma-squeeze circuit breaker.
- **Bryzgalova, Pavlova & Sikorskaya (2023, *Journal of Financial Economics*)**: retail lottery preference for cheap short-dated OTM calls persists well beyond the 2020 COVID window, into the 2021 meme-stock era and after — establishing that the mechanism this dissertation exploits is structural, not a transient pandemic artifact.
- **Hu, Kirilova, Park & Ryu (2023)**: documents the post-2022 explosion in zero-days-to-expiration (0DTE) index options volume, extending the short-dated-flow mechanism to the index level and motivating our regime-overlay risk control (Chapter 6).

Chapter 4

Signal Construction

4.1 Excess Option Volume (EOV)

$$\text{EOV}_{i,t} = \frac{1}{2} \left[\text{pctrank} \left(\frac{\text{CallVol}_{i,t}}{\text{OI}_{i,t}^{\text{call}}} \right) + \text{pctrank} \left(\frac{\text{CallVol}_{i,t}}{\text{CallVol}_{i,t}^{20d}} \right) \right] \quad (4.1)$$

4.2 Small-Lot Call Skew (SLCS)

$$\text{SLCS}_{i,t} = \frac{\text{SmallLotCallVol}_{i,t}}{\text{TotalCallVol}_{i,t}} - \frac{\text{SmallLotPutVol}_{i,t}}{\text{TotalPutVol}_{i,t}} \quad (4.2)$$

4.3 Jump-Robust Realized Volatility

Following Barndorff-Nielsen & Shephard (2004), we estimate realized volatility via bipower variation to avoid contaminating the VRP carry signal with single-day jump outliers:

$$\text{BV}_{i,t} = \frac{\pi}{2} \sum_{k=2}^n |r_{i,t,k}| |r_{i,t,k-1}| \quad (4.3)$$

Chapter 5

Backtesting Methodology

We implement a point-in-time, cost-realistic backtest with (i) a square-root market-impact model, $c_{i,t} = \eta\sigma_{i,t}\sqrt{Q_{i,t}/\text{ADV}_{i,t}}$; (ii) a borrow-fee filter excluding short candidates above a 300bp fee threshold; (iii) a turnover-penalized rebalancing rule, $w_t = w_{t-1} + (1 - \kappa)(w_t^{\text{target}} - w_{t-1})$; and (iv) Newey–West HAC-adjusted significance testing and Deflated Sharpe Ratio correction (Bailey & López de Prado, 2014) for the multiple-comparisons problem inherent in testing several candidate sub-signals.

Chapter 6

Portfolio Construction and Risk Management

6.1 Sizing

Positions are sized via half-Kelly fractional sizing, $w_i = 0.5 \cdot \hat{\mu}_i / \hat{\sigma}_i^2$, capped at 2% gross per name, with the portfolio scaled daily to a 10% annualized volatility target using a 20-day EWMA realized-volatility estimator, subject to a 3x leverage cap.

6.2 Tail Risk: The Gamma-Squeeze Circuit Breaker

Because the strategy’s carry layer is short gamma by construction (delta-hedged short straddles on rich-VRP names), we impose an intraday circuit breaker: if a held name’s intraday move exceeds 3 average-true-range units *and* its small-lot call skew z-score exceeds the 90th percentile, the position is flattened immediately, overriding the standard end-of-day rehedg schedule.

Chapter 7

Live Deployment and Monitoring

Production deployment ingests T-1 volume/OI and customer-size data pre-open, computes signal panels, generates target weights, and executes via a POV/VWAP algorithm over the first two hours of trading (deliberately avoiding the open, where the retail 0DTE flow this strategy targets is itself actively executing and spreads are elevated). A nightly CI/CD job recomputes rolling 63-day information coefficient and information ratio, triggers an automated alert if the rolling IC falls below one historical standard deviation for ten consecutive sessions, and emits a KPI/signal-decay JSON artifact, optionally forwarded to a Slack channel.

Chapter 8

Conclusion and Risk Disclosures

The retail-flow dealer-hedging mechanism documented by Barclays (2020) is theoretically grounded, has been subsequently corroborated with cleaner causal identification (Ni et al., 2021), and — per the post-2020 literature on retail options trading persistence (Bryzgalova et al., 2023) — appears structural rather than transient. The principal risk to this strategy is *crowding*: as more systematic capital targets the same dealer-hedging flow, the price impact per unit of retail order flow, and hence the strategy’s edge, should compress. We therefore recommend conservative (half-Kelly) sizing, continuous signal-decay monitoring, and treating this as a capacity-constrained (~\$300–500M) strategy rather than a scalable flagship allocation.

This document and the accompanying pipeline are internal quantitative research artifacts for research-team presentation purposes; they do not constitute investment advice and require independent compliance, legal, and risk-committee review prior to any live capital deployment.

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